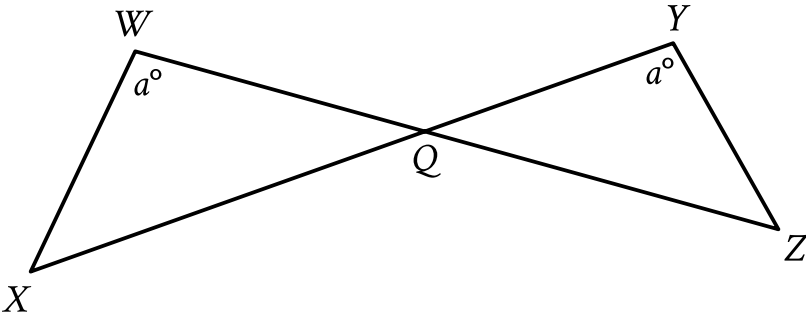


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

Correct Answer: 54

Rationale

The correct answer is **54**. The figure shown includes two triangles, triangle WQX and triangle YQZ , such that angle WQX and angle YQZ are vertical angles. It follows that angle WQX is congruent to angle YQZ . It's also given in the figure that the measures of angle W and angle Y are a° . Therefore angle W is congruent to angle Y . Since triangle WQX and triangle YQZ have two pairs of congruent angles, triangle WQX is similar to triangle YQZ by the angle-angle similarity postulate, where $\overline{YZ}$ corresponds to $\overline{WX}$, and $\overline{YQ}$ corresponds to $\overline{WQ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{WX} = \frac{YQ}{WQ}$. It's given that $YQ = 63$, $WQ = 70$, and $WX = 60$. Substituting **63** for YQ , **70** for WQ , and **60** for WX in the equation $\frac{YZ}{WX} = \frac{YQ}{WQ}$ yields $\frac{YZ}{60} = \frac{63}{70}$. Multiplying each side of this equation by **60** yields $YZ = (\frac{63}{70})(60)$, or $YZ = 54$. Therefore, the length of $\overline{YZ}$ is **54**.